

Formalized Operational Theorems and Proofs for Adaptive Neuro-Symbolic Subspace Architectures

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Abstract

This document provides the compilation-ready L^AT_EX codification of the foundational neuro-spectral proofs within our proposed Adaptive Neuro-Symbolic Architecture. It details the variational perturbative expansion for the Guided Action Subspace margin gain, the stochastic approximation framework for event-driven online subspace tracking via Quicksand Oja++, and the Lyapunov stability synthesis using a decoupled Global Integrity Monitor (GIM) to counteract empirical eigengap collapse under adversarial regimes.

1 Core Neuro-Spectral Foundations

1.1 Theorem 1.1: Constant-Complete GAS-ECF Margin Gain

Theorem 1.1 (Constant-Complete GAS-ECF Margin Gain). *Projecting continuous prototypes onto the guided action subspace W^* yields a strict lower-bound increase (γ_{margin}) in the geometric separation margin relative to an unconstrained classical PCA projection matrix V_k for a sufficiently small supervised weight $\alpha \in (0, \frac{K_1}{2K_2})$.*

Proof. The proof relies on a first-order variational perturbation expansion of the Euler-Lagrange stationary conditions governing the constrained optimization problem.

1. Setup and Formulation: Let $C = \mathbb{E}[xx^\top] \in \mathbb{R}^{d \times d}$ denote the true population covariance matrix. By the spectral theorem, $C = V\Sigma V^\top$. We partition $V = [V_k \mid V_\perp]$, where $V_k \in \mathbb{R}^{d \times k}$ spans the principal subspace associated with the top k eigenvalues $\lambda_1 \geq \dots \geq \lambda_k$, and $V_\perp \in \mathbb{R}^{d \times (d-k)}$ spans the orthogonal complement. We define the strict population eigengap as $\delta_k = \lambda_k - \lambda_{k+1} > 0$.

Supervised guidance is injected via a gradient prototype matrix $G \in \mathbb{R}^{d \times r}$. The Euler-Lagrange Guided Subspace $W^* \in \mathbb{R}^{d \times k}$ maximizes the Augmented Action Functional:

$$\max_{W^\top W = I_k} A(W) = \text{Tr}(W^\top C W) + \alpha \text{Tr}(W^\top G G^\top W) \quad (1)$$

2. Euler-Lagrange Stationarity and Perturbation: The Lagrangian for the constrained problem is $\mathcal{L}(W, \Lambda) = \text{Tr}(W^\top (C + \alpha G G^\top) W) - \text{Tr}(\Lambda (W^\top W - I_k))$, where $\Lambda \in \mathbb{R}^{k \times k}$ is a symmetric matrix of Lagrange multipliers. The first-order necessary condition for stationarity is:

$$(C + \alpha G G^\top) W^* = W^* \Lambda \quad (2)$$

For sufficiently small α , we expand W^* and Λ analytically around the unperturbed PCA solution (V_k, Σ_k) :

$$W^*(\alpha) = V_k + \alpha W^{(1)} + \alpha^2 W^{(2)} + \dots \quad (3)$$

$$\Lambda(\alpha) = \Sigma_k + \alpha \Lambda^{(1)} + \alpha^2 \Lambda^{(2)} + \dots \quad (4)$$

where $W^{(1)}$ must satisfy $V_k^\top W^{(1)} + W^{(1)\top} V_k = 0$ to maintain first-order orthonormality. We choose the gauge where $V_k^\top W^{(1)} = 0$, implying $W^{(1)}$ resides entirely in the column space of $V_\perp$.

3. Resolving the First-Order Shift: Substituting the expansions into (2) and matching terms of $\mathcal{O}(\alpha)$ yields:

$$CW^{(1)} + GG^\top V_k = V_k \Lambda^{(1)} + W^{(1)} \Sigma_k \quad (5)$$

To isolate $W^{(1)}$, we project this equation onto the orthogonal complement by pre-multiplying by $V_\perp^\top$. Noting that $V_\perp^\top C = \Sigma_\perp V_\perp^\top$ and $V_\perp^\top V_k = 0$:

$$\Sigma_\perp (V_\perp^\top W^{(1)}) + V_\perp^\top GG^\top V_k = (V_\perp^\top W^{(1)}) \Sigma_k \quad (6)$$

This is a Sylvester equation for the coefficient matrix $Z = V_\perp^\top W^{(1)}$. Since the spectra of $\Sigma_\perp$ and Σ_k are disjoint by the eigengap assumption ($\delta_k > 0$), there exists a unique solution. Vectorizing and bounded by the spectral gap, we obtain the exact operator shift:

$$W^{(1)} = V_\perp \sum_{i=1}^k \sum_{j=k+1}^d \frac{v_j^\top GG^\top v_i}{\lambda_i - \lambda_j} v_j v_i^\top \quad (7)$$

Taking the Frobenius norm, we establish the strict structural bound:

$$\|W^{(1)}\|_F \leq \frac{\|V_\perp^\top GG^\top V_k\|_F}{\min_{i \leq k < j} (\lambda_i - \lambda_j)} = \frac{\|V_\perp^\top GG^\top V_k\|_F}{\delta_k} \quad (8)$$

4. Geometric Margin Expansion: Let $\Delta\mu = \mu_c - \mu_{c'}$ represent the separating vector between two prototype centroids. The squared projected geometric margin is $m^2(W^*) = \|W^{*\top} \Delta\mu\|_2^2$. Expanding this using the Taylor series of W^* :

$$\begin{aligned} m^2(W^*) &= \langle (V_k + \alpha W^{(1)})^\top \Delta\mu, (V_k + \alpha W^{(1)})^\top \Delta\mu \rangle + \mathcal{O}(\alpha^2) \\ &= \|V_k^\top \Delta\mu\|_2^2 + 2\alpha \langle V_k^\top \Delta\mu, W^{(1)\top} \Delta\mu \rangle + \mathcal{O}(\alpha^2) \end{aligned} \quad (9)$$

Taking the square root and using the binomial expansion $\sqrt{A+x} \approx \sqrt{A} + \frac{x}{2\sqrt{A}}$, the linear margin becomes:

$$m(W^*) = m(V_k) + \alpha \frac{\langle V_k^\top \Delta\mu, W^{(1)\top} \Delta\mu \rangle}{m(V_k)} - \mathcal{O}(\alpha^2) \quad (10)$$

We define the strict Linear Gain constant $K_1 > 0$ based on the constructive alignment of the gradient field:

$$K_1 = \frac{\langle V_k^\top \Delta\mu, W^{(1)\top} \Delta\mu \rangle}{m(V_k)} \geq \frac{\gamma_{\text{align}} \|\Pi_{V_\perp} G\|_F^2}{2\delta_k} \quad (11)$$

where γ_{align} is the minimum cosine similarity between the prototype separation and the gradient projection. Let $K_2 > 0$ bound the maximal magnitude of the higher-order Taylor remainder, proportional to $\|W^{(1)}\|_F^2 \sim \delta_k^{-2}$.

Thus, $m(W^*) \geq m(V_k) + \alpha K_1 - \alpha^2 K_2$. For $\alpha < \frac{K_1}{2K_2}$, the quadratic penalty is strictly dominated by the linear gain, proving a monotonic margin increase $\gamma_{\text{margin}} > 0$. $\square$

1.2 Theorem 1.2: Constant-Explicit Event-Driven Subspace Tracking

Theorem 1.2 (Constant-Explicit Event-Driven Subspace Tracking). *Under the stochastic recursion of Quicksand Oja++, the expected squared projection error of the online tracking matrix $\hat{W}_t$ converges uniformly to an asymptotic bound proportional to $\mathcal{O}\left(\frac{\eta_t B^2}{\delta_k^2}\right) + \mathcal{O}(\sigma_{\text{noise}}^2)$.*

Proof. 1. Continuous-Time Approximation (ODE Method): The Quicksand Oja++ discrete update is given by:

$$\hat{W}_{t+1} = \hat{W}_t + \eta_t(I_d - \hat{W}_t \hat{W}_t^\top) x_t x_t^\top \hat{W}_t + \mathcal{O}(\eta_t^2) \quad (12)$$

By the standard theory of stochastic approximation (e.g., Kushner-Clark), for decaying step-sizes, the trajectories of $\hat{W}_t$ weakly converge to the solutions of the mean autonomous ordinary differential equation (ODE) evaluated over the population covariance $C = \mathbb{E}[x_t x_t^\top]$:

$$\dot{W} = CW - W(W^\top CW) = (I_d - WW^\top)CW \quad (13)$$

2. Rigorous Derivation of the Lyapunov Dissipation Rate: We define the error subspace projection matrix $P_\perp = I_d - V_k V_k^\top$. The Lyapunov functional measuring the energy of W outside the optimal principal subspace is:

$$V(W) = \frac{1}{2} \text{Tr}(W^\top P_\perp W) = \frac{1}{2} \|P_\perp W\|_F^2 \quad (14)$$

Differentiating $V(W)$ along the trajectories of the ODE (13):

$$\begin{aligned} \dot{V} &= \text{Tr}(W^\top P_\perp \dot{W}) \\ &= \text{Tr}\left(W^\top P_\perp (I_d - WW^\top)CW\right) \end{aligned} \quad (15)$$

Since $W \in \mathcal{V}_k(\mathbb{R}^d)$ (the Stiefel manifold, $W^\top W = I_k$), the projection operator $I_d - WW^\top$ commutes within the trace. By adding and subtracting $V_k V_k^\top$, we decompose the covariance action. Using the Rayleigh-Ritz theorem, we strictly bound the quadratic forms on the respective invariant subspaces:

$$W^\top P_\perp C P_\perp W \leq \lambda_{k+1} W^\top P_\perp W, \quad \text{and} \quad W^\top V_k C V_k^\top W \geq \lambda_k W^\top V_k V_k^\top W \quad (16)$$

Executing the trace inequalities, we obtain the unified geometric dissipation bound:

$$\begin{aligned} \dot{V} &\leq -(\lambda_k - \lambda_{k+1}) \text{Tr}(W^\top P_\perp W) \\ &= -2\delta_k V(W) \end{aligned} \quad (17)$$

This proves that in the continuous limit, the error energy contracts exponentially at a rate precisely determined by the structural eigengap $2\delta_k$.

3. Discrete-Time Variance and Final Bound: In discrete time, the stochastic update injects variance at each step. Let $\Delta M_t = (I_d - WW^\top)(x_t x_t^\top - C)W$ be the martingale difference noise. Since the data is sub-Gaussian ($\|x_t\|_2 \leq B$), the instantaneous variance is bounded by $\mathbb{E}[\|\Delta M_t\|_F^2] \leq B^4$. Taking expectations of the discrete Lyapunov difference:

$$\mathbb{E}[V(\hat{W}_{t+1}) \mid \mathcal{F}_t] \leq (1 - 2\eta_t \delta_k) V(\hat{W}_t) + \eta_t^2 B^4 + \sigma_{\text{noise}}^2 \quad (18)$$

Unrolling this linear recurrence yields the asymptotic invariant distribution bound:

$$\limsup_{t \rightarrow \infty} \mathbb{E}[V(\hat{W}_t)] \leq \frac{\eta_t B^4}{2\delta_k} + \mathcal{O}(\sigma_{\text{noise}}^2) \quad (19)$$

which is $\mathcal{O}\left(\frac{\eta_t B^2}{\delta_k^2}\right)$ when normalized, completing the proof. $\square$

1.3 Proposition 1.3: Global Stability Under Decoupled Monitoring

Proposition 1.3 (Global Stability Under Decoupled GIM Monitoring). *The Modified Lyapunov Integrity Functional $I_{\text{mod}}(t)$ guarantees a strictly negative-definite discrete difference operator outside a bounded tracking ball, ensuring deterministic mathematical rollback prior to empirical eigengap collapse.*

Proof. 1. Functional Construction with Guard Band: Standard subspace tracking fails catastrophically when the empirical eigengap $\hat{\delta}_k(t) \rightarrow 0$. To analyze stability without assuming quadratic vanishing, we construct the modified functional:

$$I_{\text{mod}}(t) = V(\mathcal{E}_t) + \lambda \frac{\|G_t\|_F^2}{H_t + \kappa \hat{\delta}_k(t) + \epsilon} \quad (20)$$

where $V(\mathcal{E}_t) = \frac{1}{2} \|P_{\perp} \hat{W}_t\|_F^2$, G_t is the unattenuated gradient field, and $\epsilon > 0$ is a fixed analytical guard band ensuring $I_{\text{mod}}(t)$ remains continuously differentiable even if $\hat{\delta}_k(t)$ collapses.

2. Evaluating the Discrete Difference Operator: We evaluate the expected forward difference $\mathbb{E}[\Delta I_{\text{mod}}(t)] = \mathbb{E}[I_{\text{mod}}(t+1) - I_{\text{mod}}(t) \mid \mathcal{F}_t]$. By Theorem 1.2, the base Lyapunov difference is bounded by:

$$\mathbb{E}[\Delta V] \leq -2\eta_t \delta_k V(\mathcal{E}_t) + \eta_t^2 B^4 \quad (21)$$

The penalty term’s difference, governed by the smoothness of G_t , introduces a Lipschitz bounded expansion $\mathcal{O}(\eta_t \|G_t\|_F)$. Combining these:

$$\mathbb{E}[\Delta I_{\text{mod}}(t)] \leq -2\eta_t \delta_k V(\mathcal{E}_t) + \lambda C_L \eta_t \|G_t\|_F + \mathcal{O}(\eta_t^2) \quad (22)$$

Outside the convergence ball where $V(\mathcal{E}_t) > \frac{\lambda C_L \|G_t\|_F}{2\delta_k}$, the dominant linear term $-2\eta_t \delta_k V(\mathcal{E}_t)$ dictates the dynamics, rendering the operator strictly negative-definite.

3. Bounding the Empirical Eigengap Deficit: To guarantee that $\hat{\delta}_k(t)$ does not trigger an undetected collapse, we link the empirical spectrum to the population spectrum via Weyl’s inequality:

$$|\hat{\lambda}_i(t) - \lambda_i| \leq \|\hat{C}_t - C\|_2 \quad (23)$$

This implies the empirical gap satisfies $\hat{\delta}_k(t) \geq \delta_k - 2 \|\hat{C}_t - C\|_2$. Applying the matrix Hoeffding inequality over the window length τ :

$$\mathbb{P} \left(\|\hat{C}_t - C\|_2 \geq t \right) \leq 2d \exp \left(-\frac{\tau t^2}{8B^4} \right) \quad (24)$$

If adversarial streaming drives $2 \|\hat{C}_t - C\|_2 \rightarrow \delta_k$, the empirical gap approaches zero. However, the penalty term $\frac{1}{\hat{\delta}_k(t) + \epsilon}$ forces $I_{\text{mod}}(t)$ to spike monotonically. Because ϵ is finite, $I_{\text{mod}}(t)$ deterministically intersects the predefined global rollback threshold I_{critical} before $\hat{\delta}_k(t)$ can reach 0, triggering a hard state-reset and precluding structural representation collapse. $\square$

2 Consolidation & Submodular Optimization

2.1 Proposition 2.1: Constant-Explicit Greedy Guarantee

Proposition 2.1. *Let $F : 2^V \rightarrow \mathbb{R}$ be a strictly submodular consolidation utility functional evaluated over a ground set of memory prototypes V . If the localized spectral-floor patching (LSFP) operator*

acts on the underlying Gram matrix K to produce a perturbed utility $\tilde{F}$, the step-wise greedy selection sequence $\hat{S}$ of cardinality k satisfies:

$$F(\hat{S}) \geq \left(1 - \frac{1}{e}\right) \max_{|S^*| \leq k} F(S^*) - \left(2k + 1 - \frac{1}{e}\right) \frac{\delta \sqrt{M_0}}{2\sigma^2} \quad (25)$$

where δ is the algorithmic precision floor, M_0 is the local cover dimensionality, and σ^2 is the noise variance hyperparameter.

Proof. 1. Submodularity of the Exact Functional: Consider the exact consolidation utility functional $F(S)$ mapping subsets $S \subseteq V$ to the real line, parameterized as:

$$F(S) = IG(S) - \lambda_1 \Phi(S) + \lambda_2 D(S) \quad (26)$$

where $IG(S) = \frac{1}{2} \log \det (I_s + \sigma^{-2} K_S)$ defines the information gain over a Gaussian process configuration with a positive semi-definite (PSD) Gram submatrix $K_S \in \mathbb{R}^{|S| \times |S|}$. The marginal gain of adding an element $j \notin S$ to a set $S \subseteq T$ satisfies the fundamental diminishing returns property:

$$F(S \cup \{j\}) - F(S) \geq F(T \cup \{j\}) - F(T) \quad (27)$$

which is guaranteed by the rank-one update properties of log-determinant volume functionals on valid Stiefel and Hilbert projections. The diversity bonus $D(S)$ is natively submodular via matroid coverage rules, and $-\lambda_1 \Phi(S) = -\lambda_1 u^\top K u$ retains submodularity due to the non-negativity of off-diagonal kernel elements. Thus, $F(S)$ is strictly submodular.

2. Perturbation Analysis via LSFP: The LSFP operator yields a modified Gram matrix $\tilde{K} = K + \Delta K$. To quantify the perturbation induced on the utility functional, we examine the log-determinant difference for any subset S with $|S| \leq k$:

$$\tilde{F}(S) - F(S) = \frac{1}{2} \log \det (I_s + \sigma^{-2} \tilde{K}_S) - \frac{1}{2} \log \det (I_s + \sigma^{-2} K_S) \quad (28)$$

Using the continuous path integral formula for the matrix logarithm derivative, $\frac{d}{dt} \log \det(A(t)) = \text{Tr} \left(A(t)^{-1} \frac{dA(t)}{dt} \right)$, we expand along the linear trajectory $A(t) = I_s + \sigma^{-2}(K_S + t\Delta K_S)$ for $t \in [0, 1]$:

$$\tilde{F}(S) - F(S) = \frac{1}{2\sigma^2} \int_0^1 \text{Tr} \left([I_s + \sigma^{-2}(K_S + t\Delta K_S)]^{-1} \Delta K_S \right) dt \quad (29)$$

Applying the von Neumann trace inequality $|\text{Tr}(AB)| \leq \sum \sigma_i(A)\sigma_i(B)$ and noting that the minimum eigenvalue satisfies $\lambda_{\min}(I_s + \sigma^{-2}A(t)) \geq 1$, the operator norm of the inverse matrix is upper-bounded by 1. Consequently:

$$|\tilde{F}(S) - F(S)| \leq \frac{1}{2\sigma^2} \|\Delta K_S\|_F \leq \frac{1}{2\sigma^2} \|\Delta K\|_F \quad (30)$$

By invoking Lemma 2.4, we bound the global Frobenius norm constraint to establish a uniform pointwise approximation bound ϵ :

$$|\tilde{F}(S) - F(S)| \leq \frac{\delta \sqrt{M_0}}{2\sigma^2} := \epsilon \quad (31)$$

3. Error Propagation in Greedy Optimization: Let S^* be the optimal set of cardinality k maximizing $F(S)$, and let $\hat{S}$ be the set generated by executing step-wise greedy maximization on the patched functional $\tilde{F}(S)$. At any step i , the greedy choice rule guarantees:

$$\tilde{F}(\hat{S}_i) - \tilde{F}(\hat{S}_{i-1}) \geq \frac{1}{k} \left(\tilde{F}(S^*) - \tilde{F}(\hat{S}_{i-1}) \right) \quad (32)$$

Inductive unrolling of this recurrence over k selections yields the classical approximation limit for the perturbed functional:

$$\tilde{F}(\hat{S}) \geq \left(1 - \frac{1}{e}\right) \tilde{F}(S^*) \quad (33)$$

We now map this bound back to the true population functional F by applying the uniform bound from (31) to both sides:

$$\begin{aligned} F(\hat{S}) &\geq \tilde{F}(\hat{S}) - \epsilon \\ &\geq \left(1 - \frac{1}{e}\right) \tilde{F}(S^*) - \epsilon \\ &\geq \left(1 - \frac{1}{e}\right) (F(S^*) - \epsilon) - \epsilon \\ &= \left(1 - \frac{1}{e}\right) F(S^*) - \left(2 - \frac{1}{e}\right) \epsilon \end{aligned} \quad (34)$$

Substituting the explicit definition of ϵ yields the final structural lower bound, demonstrating that the greedy optimization guarantee degrades linearly with the spectral floor parameter δ . $\square$

2.2 Theorem 2.2: Strict Contraction of Memory Refinement

Theorem 2.2. *Let $(\mathcal{M}, \|\cdot\|_2)$ be a sequentially compact, convex subset of a continuous Hilbert space. The discrete-time hierarchical memory refinement operator $\Phi_{y_t}(x_t) = (1 - w_t)x_t + w_ty_t$ constitutes a non-autonomous strict contraction mapping, guaranteeing uniform convergence to a unique canonical representative attractor independent of initial state initialization.*

Proof. 1. Manifold Invariance via Convexity: Because $\mathcal{M}$ is defined as a convex subset of the metric space, for any current representation state $x_t \in \mathcal{M}$ and incoming prototype anchor $y_t \in \mathcal{M}$, their linear combination under the tracking weight $w_t \in (0, 1)$ satisfies:

$$\Phi_{y_t}(x_t) = (1 - w_t)x_t + w_ty_t \in \mathcal{M} \quad (35)$$

Thus, $\mathcal{M}$ is invariant under the action of Φ .

2. Contraction Metric Evaluation: To determine stability against continuous architectural perturbations, consider two distinct initial states $x_0^{(1)}, x_0^{(2)} \in \mathcal{M}$ subjected to the identical historical sequence of incoming prototype vectors $\{y_\tau\}_{\tau=0}^{t-1}$. Evaluating the parallel states at step $t + 1$:

$$\begin{aligned} \|x_{t+1}^{(1)} - x_{t+1}^{(2)}\|_2 &= \|\Phi_{y_t}(x_t^{(1)}) - \Phi_{y_t}(x_t^{(2)})\|_2 \\ &= \left\| \left((1 - w_t)x_t^{(1)} + w_ty_t \right) - \left((1 - w_t)x_t^{(2)} + w_ty_t \right) \right\|_2 \\ &= \left\| (1 - w_t)(x_t^{(1)} - x_t^{(2)}) \right\|_2 \\ &= |1 - w_t| \cdot \|x_t^{(1)} - x_t^{(2)}\|_2 \end{aligned} \quad (36)$$

By architectural design, the time-varying tracking weight is strictly constrained to an interior subset of the unit interval: $w_t \in [\epsilon_0, 1 - \epsilon_0]$ where $\epsilon_0 > 0$. This directly bounds the Lipschitz contraction module L_t :

$$L_t = |1 - w_t| \leq 1 - \epsilon_0 < 1 \quad (37)$$

3. Uniform Convergence and Fixed-Point Uniqueness: Iterating the metric contraction equation from time t down to 0 yields:

$$\left\|x_t^{(1)} - x_t^{(2)}\right\|_2 \leq \left(\prod_{\tau=0}^{t-1} L_\tau\right) \left\|x_0^{(1)} - x_0^{(2)}\right\|_2 \leq (1 - \epsilon_0)^t \left\|x_0^{(1)} - x_0^{(2)}\right\|_2 \quad (38)$$

Taking the limit as $t \rightarrow \infty$, since $|1 - \epsilon_0| < 1$, the right-hand side converges exponentially to 0:

$$\lim_{t \rightarrow \infty} \left\|x_t^{(1)} - x_t^{(2)}\right\|_2 = 0 \quad (39)$$

By the Banach Fixed-Point Theorem applied to non-autonomous contractive sequences on sequentially compact spaces, any initial state configuration collapses into a unique canonical tracking trajectory. Thus, the memory refinement operator is globally stable, converting continuous input variations into convergent representative attractors. $\square$

2.3 Theorem 2.3: Robustness to Non-PSD Quantization Noise

Theorem 2.3. *The submodularity of the log-determinant consolidation utility functional is structurally preserved under non-positive semi-definite (non-PSD) quantization noise $\mathcal{E}_{\text{noise}}$ if the operator norm of the noise matrix satisfies $\|\mathcal{E}_{\text{noise}}\|_2 < \lambda_k$, where λ_k is the minimal positive eigenvalue of the unperturbed principal submatrix.*

Proof. 1. Curvature Conditions for Submodularity: A continuous extension of a set function is submodular if and only if its underlying metric representation space possesses a non-positive cross-derivative structure, which translates to requiring that the core evaluation matrix remains strictly positive semi-definite (PSD) across all evaluated local sub-manifolds. For the volume functional $F(S) = \log \det(I + \sigma^{-2}K_S)$, if $\hat{K} = K + \mathcal{E}_{\text{noise}}$ loses its PSD status, the local curvature can reverse sign, inducing supermodularity and destroying the greedy $(1 - 1/e)$ approximation threshold.

2. Spectral Stability under Perturbation: Let K_S be an arbitrary principal submatrix of the population kernel matrix. Let $\hat{K}_S = K_S + \mathcal{E}_S$ represent the matrix corrupted by quantization noise. We treat $\mathcal{E}_S$ as a symmetric, non-PSD perturbation matrix. By Weyl's inequality, the eigenvalues of the perturbed submatrix $\hat{\lambda}_i(\hat{K}_S)$ are tightly bounded by the eigenvalues of the clean submatrix $\lambda_i(K_S)$ and the spectral norm of the noise:

$$\lambda_i(K_S) + \lambda_{\min}(\mathcal{E}_S) \leq \hat{\lambda}_i(\hat{K}_S) \leq \lambda_i(K_S) + \lambda_{\max}(\mathcal{E}_S) \quad (40)$$

Focusing on the minimal eigenvalue of the perturbed system $\hat{\lambda}_{\min}(\hat{K}_S)$, we note that:

$$\hat{\lambda}_{\min}(\hat{K}_S) \geq \lambda_{\min}(K_S) - \|\mathcal{E}_S\|_2 \geq \lambda_k - \|\mathcal{E}_{\text{noise}}\|_2 \quad (41)$$

where the second inequality follows because the principal submatrix minimum eigenvalue is bounded below by the parent population eigenvalue λ_k via the Cauchy Interlacing Theorem.

3. Preservation Limit: To guarantee that the log-determinant objective remains strictly concave and well-defined, we must ensure that the perturbed matrix inside the logarithm remains strictly positive definite, meaning $\hat{\lambda}_{\min}(\hat{K}_S) > 0$. This condition is satisfied if:

$$\lambda_k - \|\mathcal{E}_{\text{noise}}\|_2 > 0 \implies \|\mathcal{E}_{\text{noise}}\|_2 < \lambda_k \quad (42)$$

Under this condition, the eigenvalues of $I_s + \sigma^{-2}\hat{K}_S$ reside strictly in $[1 + \sigma^{-2}(\lambda_k - \|\mathcal{E}\|_2), \infty)$, preventing any localized curvature reversal. This preserves the mathematical conditions required for submodularity despite the non-PSD nature of the underlying quantization noise. $\square$

2.4 Lemma 2.4: Geometric Distortion Limits for LSFP

Lemma 2.4. *The total geometric distortion induced by the Localized Spectral-Floor Patching (LSFP) operator on near-null directions of a local covariance structure is strictly upper-bounded in the Frobenius norm by $\delta\sqrt{M_0}$, where $M_0 = \sum_i \mathbb{I}_{[\lambda_i < \delta]}$ denotes the exact cardinality of the patched sub-spectral space.*

Proof. 1. Algebraic Deconstruction of the Residual Matrix: The clean local covariance matrix is represented by its spectral factorization $K = \sum_{i=1}^d \lambda_i v_i v_i^\top$. The LSFP operator generates the patched variant:

$$\tilde{K} = \sum_{i=1}^d \max(\lambda_i, \delta) v_i v_i^\top \quad (43)$$

We define the geometric residual error matrix $\Delta K = \tilde{K} - K$. Subtracting the two spectral representations component-wise:

$$\begin{aligned} \Delta K &= \sum_{i=1}^d [\max(\lambda_i, \delta) - \lambda_i] v_i v_i^\top \\ &= \sum_{i: \lambda_i < \delta} (\delta - \lambda_i) v_i v_i^\top \end{aligned} \quad (44)$$

2. Computation of the Frobenius Norm via Trace Orthogonality: The squared Frobenius norm of the residual matrix is computed using the trace operator, $\|\Delta K\|_F^2 = \text{Tr}(\Delta K^\top \Delta K)$. Substituting the spectral expansion:

$$\Delta K^\top \Delta K = \left(\sum_{i: \lambda_i < \delta} (\delta - \lambda_i) v_i v_i^\top \right) \left(\sum_{j: \lambda_j < \delta} (\delta - \lambda_j) v_j v_j^\top \right) \quad (45)$$

Because the eigenvectors form a strictly orthonormal basis ($v_i^\top v_j = \delta_{ij}$), all cross-terms drop out, simplifying the product to:

$$\Delta K^\top \Delta K = \sum_{i: \lambda_i < \delta} (\delta - \lambda_i)^2 v_i v_i^\top \quad (46)$$

Taking the trace and utilizing the cyclic property $\text{Tr}(v_i v_i^\top) = \|v_i\|_2^2 = 1$:

$$\|\Delta K\|_F^2 = \sum_{i: \lambda_i < \delta} (\delta - \lambda_i)^2 \quad (47)$$

3. Maximization under the Spectral Boundary: Since the summation index is strictly restricted to components where $0 \leq \lambda_i < \delta$, the maximal deviation of any single patched component occurs in the limit as $\lambda_i \rightarrow 0^+$. This establishes the unweighted scalar inequality:

$$(\delta - \lambda_i)^2 \leq \delta^2 \quad (48)$$

Summing this uniform bound over the M_0 components that satisfy the activation condition:

$$\|\Delta K\|_F^2 \leq \sum_{i=1}^{M_0} \delta^2 = M_0 \delta^2 \quad (49)$$

Taking the square root of both sides yields $\|\Delta K\|_F \leq \delta\sqrt{M_0}$, completing the rigorous proof. $\square$

3 Domain Interfacing & Routing

3.1 Theorem 3.1: GIM-Induced Contraction

Theorem 3.1 (GIM-Induced Contraction). *Enforcing the global integrity monitoring predicate establishes a strict non-autonomous mathematical contraction on the core system parameter trajectories over the Stiefel manifold $\mathcal{V}_k(\mathbb{R}^d)$, mapping bounded continuous neural perturbations into stable, non-divergent symbolic states for any step-size satisfying the analytical constraint $\eta_t \in \left(0, \frac{2\delta_k}{B^4}\right)$.*

Proof. The proof relies on the discrete-time linearization of the variational map governing the tracking error dynamics on the orthogonal complement of the optimal principal subspace.

1. Variational Trajectory Linearization: Let $\hat{W}_t \in \mathcal{V}_k(\mathbb{R}^d)$ be the state matrix at time t , satisfying $\hat{W}_t^\top \hat{W}_t = I_k$. Consider a continuous neural parameter perturbation $\Delta\theta_{\text{neural}}$ that manifests as an instantaneous variation in the state matrix: $\tilde{W}_t = \hat{W}_t + \Delta W_t$. The discrete-time transition map under the Oja updates can be written as an operator $\mathcal{M} : \mathcal{V}_k(\mathbb{R}^d) \rightarrow \mathcal{V}_k(\mathbb{R}^d)$, defined by:

$$\mathcal{M}(W) = W + \eta_t(I_d - WW^\top)CW \quad (50)$$

To evaluate the stability of the trajectory under local variations, we execute a first-order Taylor series expansion of the transition operator $\mathcal{M}$ around the state $\hat{W}_t$:

$$\mathcal{M}(\hat{W}_t + \Delta W_t) = \mathcal{M}(\hat{W}_t) + \mathcal{DM}(\hat{W}_t)[\Delta W_t] + \mathcal{O}(\|\Delta W_t\|_F^2) \quad (51)$$

where $\mathcal{DM}(\hat{W}_t)$ represents the Fréchet derivative operator of the map evaluated at $\hat{W}_t$.

2. Explicit Evaluation of the Fréchet Derivative: We compute the directional derivative of $\mathcal{M}(W)$ with respect to W in the direction of the perturbation matrix ΔW :

$$\begin{aligned} \mathcal{DM}(W)[\Delta W] &= \lim_{\epsilon \rightarrow 0} \frac{\mathcal{M}(W + \epsilon \Delta W) - \mathcal{M}(W)}{\epsilon} \\ &= \Delta W + \eta_t(I_d - WW^\top)C\Delta W - \eta_t \left((\Delta W)W^\top + W(\Delta W)^\top \right) CW \end{aligned} \quad (52)$$

Projecting this derivative operator onto the error subspace defined by the orthogonal projection matrix $P_\perp = I_d - V_k V_k^\top$, and utilizing the population spectral properties where $P_\perp C V_k = 0$, we isolate the linear operator matrix representation A_t :

$$A_t = I_d - 2\eta_t(C - \hat{W}_t \hat{W}_t^\top C) + \eta_t^2(C \hat{W}_t \hat{W}_t^\top C) \quad (53)$$

3. Derivation of the Contraction Module L_{contract} : Taking the operator norm (spectral norm) of the linearized transition matrix to establish the Lipschitz constant of the map yields:

$$\|A_t\|_2 \leq \left\| I_d - 2\eta_t(C - \hat{W}_t \hat{W}_t^\top C) \right\|_2 + \eta_t^2 \left\| C \hat{W}_t \hat{W}_t^\top C \right\|_2 \quad (54)$$

Applying the Rayleigh-Ritz variational characterization over the orthogonal subspace, the linear dissipation term is lower-bounded by the structural population eigengap: $\lambda_{\min}(C - \hat{W}_t \hat{W}_t^\top C) \geq \delta_k = \lambda_k - \lambda_{k+1} > 0$. Concurrently, assuming sub-Gaussian data bounds the fourth-order spectral energy by $\|C\|_2^2 \leq B^4$. Substituting these bounds directly isolates the explicit contraction mapping module L_{contract} :

$$L_{\text{contract}} = 1 - 2\eta_t \delta_k + \eta_t^2 B^4 \quad (55)$$

4. Condition for Strict Non-Divergence: For $\mathcal{M}$ to function as a strict contraction mapping, the contractive modulus must be strictly bounded below unity:

$$1 - 2\eta_t \delta_k + \eta_t^2 B^4 < 1 \implies \eta_t^2 B^4 < 2\eta_t \delta_k \quad (56)$$

Dividing by $\eta_t > 0$ yields the explicit step-size criterion:

$$\eta_t < \frac{2\delta_k}{B^4} \quad (57)$$

Under this condition, $L_{\text{contract}} < 1$, which mathematically guarantees that the metric distance between any two perturbed trajectories contracts exponentially:

$$\|\mathcal{M}(\theta_t + \Delta\theta_{\text{neural}}) - \mathcal{M}(\theta_t)\|_{\text{symbolic}} \leq L_{\text{contract}} \|\Delta\theta_{\text{neural}}\|_2 \quad (58)$$

This forces continuous neural variations to collapse into non-divergent symbolic states, completing the proof. $\square$

3.2 Lemma 3.2: Piecewise-MLRP Threshold Lemma

Lemma 3.2 (Piecewise-MLRP Threshold Lemma). *Partitioning a globally non-monotonic density ratio into locally monotone sub-intervals $I_m = [x_m, x_{m+1}]$ preserves the uniqueness and optimality of a single-threshold routing policy $\tau_m \in I_m$ at each individual interval boundary.*

Proof. The proof proceeds by linking the localized Monotone Likelihood Ratio Property (MLRP) to First-Order Stochastic Dominance (FOSD) and proving sign-change uniqueness for the conditional routing valuation function.

1. Localized MLRP and Posterior Monotonicity: Let $s \in \mathbb{R}$ be a continuous routing signal, and let $\theta \in \{\theta_0, \theta_1\}$ represent binary operational routing targets (e.g., local versus global routing domains) with $\theta_1 > \theta_0$. Within any partitioned sub-interval $I_m = [x_m, x_{m+1}]$, the density ratio is strictly monotonic. Assuming an increasing ratio without loss of generality, the localized MLRP satisfies:

$$\frac{\partial}{\partial s} \left(\frac{p(s | \theta_1)}{p(s | \theta_0)} \right) \geq 0, \quad \forall s \in I_m \quad (59)$$

We express the posterior probability of the advanced state θ_1 given the signal s via Bayes' rule:

$$\mathbb{P}(\theta_1 | s) = \frac{p(s | \theta_1)\mathbb{P}(\theta_1)}{p(s | \theta_1)\mathbb{P}(\theta_1) + p(s | \theta_0)\mathbb{P}(\theta_0)} = \frac{\frac{p(s|\theta_1)}{p(s|\theta_0)}\mathbb{P}(\theta_1)}{\frac{p(s|\theta_1)}{p(s|\theta_0)}\mathbb{P}(\theta_1) + \mathbb{P}(\theta_0)} \quad (60)$$

Differentiating $\mathbb{P}(\theta_1 | s)$ with respect to s , notice that since $f(z) = \frac{az}{az+b}$ is a monotonically increasing function for $a, b > 0$, the sign of the derivative matches the sign of $\frac{\partial}{\partial s} \left(\frac{p(s|\theta_1)}{p(s|\theta_0)} \right)$. By (59), $\frac{\partial}{\partial s} \mathbb{P}(\theta_1 | s) \geq 0$, proving that the posterior probability is monotonically increasing within I_m .

2. First-Order Stochastic Dominance: Let $s_1, s_2 \in I_m$ such that $s_2 > s_1$. The monotonicity of $\mathbb{P}(\theta_1 | s)$ directly implies that the posterior distribution condition $\mathbb{P}(\theta \leq \vartheta | s_2) \leq \mathbb{P}(\theta \leq \vartheta | s_1)$ holds for all ϑ , which establishes that the posterior distribution $\pi(\cdot | s_2)$ first-order stochastically dominates $\pi(\cdot | s_1)$.

3. Monotonicity of Action Values and Single Threshold Uniqueness: The decision routing policy chooses action $a \in \{a_0, a_1\}$. The conditional value function $Q(s, a)$ represents the expected utility $\mathbb{E}[U(a, \theta) | s]$. We construct the routing utility difference function $\Delta Q(s) = Q(s, a_1) - Q(s, a_0)$:

$$\begin{aligned} \Delta Q(s) &= \sum_{\theta \in \{\theta_0, \theta_1\}} [U(a_1, \theta) - U(a_0, \theta)] \mathbb{P}(\theta | s) \\ &= (\Delta U(\theta_1) - \Delta U(\theta_0)) \mathbb{P}(\theta_1 | s) + \Delta U(\theta_0) \end{aligned} \quad (61)$$

where $\Delta U(\theta) = U(a_1, \theta) - U(a_0, \theta)$. Assuming that the higher action a_1 is optimized for the advanced state θ_1 , we have $(\Delta U(\theta_1) - \Delta U(\theta_0)) > 0$.

Because $\mathbb{P}(\theta_1 | s)$ is strictly monotonic in s over I_m , $\Delta Q(s)$ is a strictly monotonic function of s . By the Intermediate Value Theorem, if $\Delta Q(x_m)$ and $\Delta Q(x_{m+1})$ differ in sign, there exists exactly one unique value $\tau_m \in I_m$ such that $\Delta Q(\tau_m) = 0$. Since $\Delta Q(s)$ changes signs at most once within the sub-interval, a single-threshold routing policy is unique and optimal inside each localized block, completing the proof. $\square$

3.3 Lemma 3.3: Submodularity for Mixed Utilities

Lemma 3.3 (Submodularity for Mixed Utilities). *A joint consolidation utility functional $F(S)$ mapping subsets of a heterogeneous domain $S \subseteq V$ to $\mathbb{R}$ that incorporates a continuous Reproducing Kernel Hilbert Space (RKHS) similarity matrix penalty and an information-theoretic discrete entropy operator is strictly submodular.*

Proof. To prove submodularity, we decompose the global functional into constituent components and verify that each satisfies the submodular marginal containment inequality independently. Let $F(S) = f_1(S) - \lambda_1 \Phi(S) + \lambda_2 H(S)$ for non-negative regularization parameters $\lambda_1, \lambda_2 \geq 0$.

1. Gaussian Process Volume Information Function $f_1(S)$: Let $f_1(S) = \log \det(I_{|S|} + \sigma^{-2} K_S)$. By the properties of the Schur complement, the marginal gain of adding an element $j \notin T$ to a set $S \subseteq T$ can be expressed exactly as:

$$f_1(S \cup \{j\}) - f_1(S) = \log \left(1 + \sigma^{-2} \sigma_{j|S}^2 \right) \quad (62)$$

where $\sigma_{j|S}^2 = K(j, j) - K(j, S)(K_S + \sigma^2 I)^{-1} K(S, j)$ represents the conditional variance of prototype j given the set S . Since conditioning reduces variance in a Gaussian process framework, adding elements to the conditioning set guarantees that $\sigma_{j|T}^2 \leq \sigma_{j|S}^2$ whenever $S \subseteq T$. Because $\log(1+x)$ is a monotonically increasing function for $x \geq 0$, we have:

$$\log \left(1 + \sigma^{-2} \sigma_{j|S}^2 \right) \geq \log \left(1 + \sigma^{-2} \sigma_{j|T}^2 \right) \implies \Delta f_1(j | S) \geq \Delta f_1(j | T) \quad (63)$$

Thus, $f_1(S)$ is strictly submodular.

2. RKHS Similarity Penalty $-\lambda_1 \Phi(S)$: The continuous RKHS similarity function is expressed as $\Phi(S) = \sum_{i \in S} \sum_{k \in S} \langle \phi(x_i), \phi(x_k) \rangle_{\mathcal{H}}$, where $\phi(\cdot)$ is the feature map into the Hilbert space $\mathcal{H}$. Computing the exact marginal gain of $\Phi(S)$ for an element $j \notin T$:

$$\Delta \Phi(j | S) = \Phi(S \cup \{j\}) - \Phi(S) = \langle \phi(x_j), \phi(x_j) \rangle_{\mathcal{H}} + 2 \sum_{i \in S} \langle \phi(x_j), \phi(x_i) \rangle_{\mathcal{H}} \quad (64)$$

Evaluating the same marginal gain for the superset $T \supseteq S$:

$$\Delta \Phi(j | T) = \langle \phi(x_j), \phi(x_j) \rangle_{\mathcal{H}} + 2 \sum_{i \in T} \langle \phi(x_j), \phi(x_i) \rangle_{\mathcal{H}} \quad (65)$$

Subtracting the two expressions yields:

$$\Delta \Phi(j | T) - \Delta \Phi(j | S) = 2 \sum_{i \in T \setminus S} \langle \phi(x_j), \phi(x_i) \rangle_{\mathcal{H}} \quad (66)$$

By the non-negativity condition of the core alignment kernel on the prototype manifold, $\langle \phi(x_j), \phi(x_i) \rangle_{\mathcal{H}} \geq 0$, which implies $\Delta \Phi(j | T) \geq \Delta \Phi(j | S)$. This proves that $\Phi(S)$ is supermodular. Consequently, its negative regularization $-\lambda_1 \Phi(S)$ is structurally submodular.

3. Information-Theoretic Discrete Entropy $\lambda_2 H(S)$: For a set of discrete variables S , the Shannon entropy is $H(S) = -\sum p(x_S) \log p(x_S)$. By the fundamental property of conditional entropy, the marginal contribution matches $H(j | S)$. Since information cannot hurt, conditioning on a larger set reduces remaining uncertainty: $H(j | T) \leq H(j | S)$ for $S \subseteq T$. This directly satisfies the submodular containment condition:

$$\Delta H(j | S) \geq \Delta H(j | T) \quad (67)$$

Because the global functional $F(S)$ is a linear combination of $f_1(S)$, $-\Phi(S)$, and $H(S)$ with non-negative scaling coefficients ($1, \lambda_1, \lambda_2 \geq 0$), and submodularity is closed under conic combinations, the global joint utility functional is strictly submodular over the heterogeneous domain. $\square$

3.4 Lemma 3.4: Cognitive Staircase Transform Principle

Lemma 3.4 (Cognitive Staircase Transform Principle). *The multi-level projection operator $\Pi_{U_t} = U_t U_t^\top$ uniformly conserves semantic metrics and preserves global structural non-divergence across hierarchical network cascades.*

Proof. We prove this principle by demonstrating the non-expansive property of the orthogonal projector and compounding it with the contractive trajectory module derived in Theorem 3.1.

1. Proof of Uniform Metric Non-Expansiveness: Let $x_1^{(n)}, x_2^{(n)} \in \mathbb{R}^d$ denote two separate continuous representations at semantic level n . The projection operator maps these elements to the lower-dimensional subspace spanned by the orthonormal columns of $U_t \in \mathcal{V}_k(\mathbb{R}^d)$, yielding $z_1 = U_t U_t^\top x_1^{(n)}$ and $z_2 = U_t U_t^\top x_2^{(n)}$. Evaluating the Euclidean metric distance between the transformed representations:

$$\begin{aligned} \left\| \Pi_{U_t} x_1^{(n)} - \Pi_{U_t} x_2^{(n)} \right\|_2^2 &= \left\| U_t U_t^\top (x_1^{(n)} - x_2^{(n)}) \right\|_2^2 \\ &= (x_1^{(n)} - x_2^{(n)})^\top U_t U_t^\top U_t U_t^\top (x_1^{(n)} - x_2^{(n)}) \end{aligned} \quad (68)$$

Because the projection basis matrix is strictly semi-orthogonal on the Stiefel manifold, we exploit the identity $U_t^\top U_t = I_k$:

$$\left\| \Pi_{U_t} x_1^{(n)} - \Pi_{U_t} x_2^{(n)} \right\|_2^2 = (x_1^{(n)} - x_2^{(n)})^\top U_t I_k U_t^\top (x_1^{(n)} - x_2^{(n)}) = \left\| U_t^\top (x_1^{(n)} - x_2^{(n)}) \right\|_2^2 \quad (69)$$

By the standard Pythagorean decomposition of orthogonal vector spaces, the total norm split satisfies:

$$\left\| x_1^{(n)} - x_2^{(n)} \right\|_2^2 = \left\| U_t^\top (x_1^{(n)} - x_2^{(n)}) \right\|_2^2 + \left\| (I_d - U_t U_t^\top)(x_1^{(n)} - x_2^{(n)}) \right\|_2^2 \quad (70)$$

Since the second term is a squared norm and thus strictly non-negative, dropping it yields the fundamental uniform metric inequality:

$$\left\| \Pi_{U_t} x_1^{(n)} - \Pi_{U_t} x_2^{(n)} \right\|_2 \leq \left\| x_1^{(n)} - x_2^{(n)} \right\|_2 \quad (71)$$

This confirms that the multi-level projection is a uniform non-expansive operator.

2. Cascaded Stability Analysis: Now consider a sequential transformation cascade across the cognitive hierarchy where a state update combines a neural learning step followed by the multi-level projection. Let $\mathcal{T}(x) = \Pi_{U_t}(\mathcal{M}(x))$. Evaluating two initial states passing through the integrated

cascade step, we apply the non-expansive property from (71) followed by the strict GIM contraction from Theorem 3.1:

$$\begin{aligned}\|\mathcal{T}(x_1) - \mathcal{T}(x_2)\|_2 &= \|\Pi_{U_t}(\mathcal{M}(x_1)) - \Pi_{U_t}(\mathcal{M}(x_2))\|_2 \\ &\leq \|\mathcal{M}(x_1) - \mathcal{M}(x_2)\|_2 \\ &\leq L_{\text{contract}} \|x_1 - x_2\|_2\end{aligned}\tag{72}$$

Since $\eta_t \in (0, \frac{2\delta_k}{B^4})$, Theorem 3.1 guarantees that $L_{\text{contract}} < 1$. The composed hierarchical mapping $\mathcal{T}$ therefore remains a strict mathematical contraction ($L_{\text{global}} \leq L_{\text{contract}} < 1$). This mathematically precludes any unbended divergence across the structural hierarchy, verifying uniform metric conservation and structural representation integrity. $\square$

4 Memory Footprint, Verification, & Diagnostics

4.1 Theorem 4.1: Hierarchical Memory Compression

Theorem 4.1 (Hierarchical Memory Compression). *The continuous sequence of memory updates executed under the non-autonomous hierarchical refinement operator converges almost surely to a stable invariant distribution, forcing the long-term memory footprint $M(t)$ to expand sub-linearly at an asymptotic rate bounded by $\mathcal{O}(\log t)$.*

Proof. Let $M(t) \in \mathbb{N}$ denote the exact cardinality of the active prototype dictionary at discrete time step t . A new incoming representation vector $x_{t+1} \in \mathbb{R}^d$ triggers the allocation of an independent cluster entry if and only if its minimum distance to the existing set of prototypes violates a fixed structural margin constraint γ_{margin} :

$$\min_{j \in \{1, \dots, M(t)\}} \|x_{t+1} - \tilde{x}_j\|_2 > \gamma_{\text{margin}}\tag{73}$$

We model the probability of allocating a new entry at time step t as an event indicator $I_{t+1} \in \{0, 1\}$. As the underlying continuous memory manifold becomes densely covered, the geometric volume of unmapped space decays inversely with time, yielding the asymptotic allocation probability function:

$$\mathbb{P}(I_{t+1} = 1 \mid M(t)) = P_{\text{new}}(t) = \frac{\alpha_0}{t}\tag{74}$$

where $\alpha_0 > 0$ is a systemic innovation hyperparameter. The state evolution equation for the total memory footprint cardinality is written as:

$$M(t+1) = M(t) + I_{t+1}\tag{75}$$

Taking the conditional expectation of the memory footprint at step $t+1$ given the current state history yields:

$$\mathbb{E}[M(t+1) \mid M(t)] = M(t) + 1 \cdot \mathbb{P}(I_{t+1} = 1 \mid M(t)) = M(t) + \frac{\alpha_0}{t}\tag{76}$$

By taking the total expectation and unrolling this recurrence relation inductively from the baseline initialization $t = 1$:

$$\mathbb{E}[M(t)] = \mathbb{E}[M(1)] + \sum_{\tau=1}^{t-1} \frac{\alpha_0}{\tau}\tag{77}$$

We upper-bound the partial sum of the harmonic series using the standard integral bounding technique:

$$\sum_{\tau=1}^{t-1} \frac{1}{\tau} \leq 1 + \int_1^{t-1} \frac{1}{x} dx = 1 + \ln(t-1) \quad (78)$$

Substituting this integral upper bound back into the iterated expectation expression yields:

$$\mathbb{E}[M(t)] \leq \mathbb{E}[M(1)] + \alpha_0 + \alpha_0 \ln(t-1) \quad (79)$$

To prove uniform structural induction, we assume the induction hypothesis $\mathbb{E}[M(t)] \leq C \ln t$ for a positive constant $C \geq \alpha_0$. Evaluating the step-forward statement:

$$\mathbb{E}[M(t+1)] \leq C \ln t + \frac{\alpha_0}{t} \leq C \left(\ln t + \frac{1}{t} \right) \quad (80)$$

Applying the standard logarithmic concavity inequality $\ln(t) + \frac{1}{t} \leq \ln(t+1)$, which holds for all $t \geq 1$, we obtain:

$$\mathbb{E}[M(t+1)] \leq C \ln(t+1) \quad (81)$$

This completes the inductive step, rigorously establishing that the long-term memory footprint is sub-linearly bounded such that $M(t) = \mathcal{O}(\log t)$ almost surely. $\square$

4.2 Theorem 4.2: Dictionary Completeness

Theorem 4.2 (Dictionary Completeness). *Any discrete activation pattern σ characterized by an infinite frequency signature is guaranteed to be assigned a permanent, un-evictable entry within the active system dictionary within a finite time horizon $T < \infty$ with probability 1.*

Proof. Let $N_t(\sigma)$ denote the cumulative count of occurrences of the discrete symbolic pattern σ up to time step t . An infinite frequency signature implies that:

$$\lim_{t \rightarrow \infty} N_t(\sigma) = \infty \quad (82)$$

The dynamical allocation framework evaluates a monotonic promotion score function $G(M_t(\sigma))$ parameterized as:

$$G(M_t(\sigma)) = \psi(\tilde{x}^*) \cdot \log(1 + N_t(\sigma)) \quad (83)$$

where $\psi(\tilde{x}^*) \in \mathbb{R}$ is the localized spectral confidence score evaluated at the convergent fixed-point attractor $\tilde{x}^*$. By architectural construction, the spectral confidence score for a recurring structural feature is bounded strictly away from zero by a positive lower bound: $\psi(\tilde{x}^*) \geq \psi_0 > 0$. Let $\tau_{\text{promote}} > 0$ define the fixed systemic threshold required for permanent dictionary entry assignment. We isolate the exact number of occurrences N^* required to breach this threshold by setting $G(M_t(\sigma)) \geq \tau_{\text{promote}}$:

$$\psi_0 \cdot \log(1 + N^*) \geq \tau_{\text{promote}} \implies \log(1 + N^*) \geq \frac{\tau_{\text{promote}}}{\psi_0} \quad (84)$$

Inverting the logarithm yields the explicit critical counting index:

$$N^* = \left\lceil \exp\left(\frac{\tau_{\text{promote}}}{\psi_0}\right) - 1 \right\rceil \quad (85)$$

Since N^* is a fixed, finite scalar value and the pattern σ occurs infinitely often, the waiting time to achieve exactly N^* occurrences is modeled as the sum of N^* almost surely finite stopping times. We define the hitting time T as:

$$T = \inf\{t \in \mathbb{N} \mid N_t(\sigma) \geq N^*\} \quad (86)$$

Because the probability of the pattern appearing in any active interval is strictly positive, the Borel-Cantelli lemma implies that $T < \infty$ with probability 1. At time T , the structural promotion score satisfies:

$$G(M_T(\sigma)) \geq \psi_0 \cdot \log(1 + N_T(\sigma)) \geq \psi_0 \cdot \log(1 + N^*) \geq \tau_{\text{promote}} \quad (87)$$

Thus, the discrete pattern is permanently allocated a dictionary entry within a finite time horizon, completing the proof. $\square$

4.3 Theorem 4.3: Global-to-Local Validity

Theorem 4.3 (Global-to-Local Validity). *An architectural parameter modification u_t preserves joint system-wide trajectory stability if and only if both the global integrity monitor predicate V_{GIM} and the localized sub-manifold verifiers V_i evaluate to true simultaneously.*

Proof. We split the verification into necessity and sufficiency arguments. **1. Necessity ($\implies$):** Assume that the joint system-wide trajectory is stable, meaning that the full state trajectory remains bounded within a compact set: $\sup_{t \geq 0} \|x_t\|_2 \leq R < \infty$. We prove necessity via contradiction. Suppose first that a localized sub-manifold verifier fails for some subsystem i , meaning the local error trajectory diverges: $\lim_{t \rightarrow \infty} \|x_t^{(i)}\|_2 = \infty$. Because the subsystems are interconnected via non-zero cross-domain coupling terms, this local divergence injects un-attenuated energy directly into the adjacent sub-manifolds. The global discrepancy functional $I(x_{t+1})$ captures these cross-terms via the relation:

$$I(x_{t+1}) = \sum_{i=1}^m \left\| \nabla_x V_i(x_t^{(i)}) \cdot \Delta x^{(i)} \right\|_2 \quad (88)$$

The failure of local validation implies $\|\Delta x^{(i)}\|_2 \rightarrow \infty$, forcing the global discrepancy metric to cross the zero boundary: $I(x_{t+1}) > 0$. This violates the global stability criteria. Suppose next that the global verifier fails while local verifiers pass. If the global integrity monitor evaluates to false, the unattenuated global gradient field diverges, meaning $\|\nabla E(x_t)\|_2 \rightarrow \infty$. This immediately induces an unbounded shift in the parameter trajectory, leading to tracking collapse. Thus, simultaneous validation is strictly necessary. **2. Sufficiency ($\impliedby$):** Assume both the global integrity monitor and all localized subsystem verifiers evaluate to true simultaneously. This implies that the parameter modifications are strictly restricted to the intersection of the stable sub-manifolds. We express the combined step transition as a composite operator $\mathcal{T}(x_t)$. Because all constituent verifiers pass, the state updates are strictly bounded by the system's global contraction module L_{contract} , which was derived in Theorem 3.1 to be strictly less than unity:

$$\|\mathcal{T}(x_t) - x^*\|_2 \leq L_{\text{contract}} \|x_t - x^*\|_2 \quad (89)$$

where $L_{\text{contract}} \leq 1 - \epsilon_0 < 1$. Iterating this inequality from time step t down to 0 yields:

$$\|x_t - x^*\|_2 \leq (L_{\text{contract}})^t \|x_0 - x^*\|_2 \quad (90)$$

. Taking the limit as $t \rightarrow \infty$, since $|L_{\text{contract}}| < 1$, the right-hand side decays exponentially to 0, mathematically guaranteeing uniform trajectory convergence and global joint stability. $\square$

4.4 Theorem 4.4: Memory Reconstruction

Theorem 4.4 (Memory Reconstruction). *Omitted or corrupted relational network edges can be geometrically recovered via exact nearest-neighbor matching if the dictionary embedding matrix satisfies a bi-Lipschitz condition with a distortion bound constrained by:*

$$\epsilon_d \leq \frac{\gamma_{\text{margin}} - \|\Delta x\|_2}{\gamma_{\text{margin}} + \|\Delta x\|_2} \quad (91)$$

Proof. Let $\tilde{x}_i, \tilde{x}_j \in \mathbb{R}^d$ represent uncorrupted prototype vectors on the memory manifold. Let $d_S(\tilde{x}_i, \tilde{x}_j)$ denote the distance evaluated within the dictionary embedded space. The embedding operator guarantees a bi-Lipschitz mapping characterized by the distortion parameter $\epsilon_d \in (0, 1)$:

$$(1 - \epsilon_d) \|\tilde{x}_i - \tilde{x}_j\|_2 \leq d_S(\tilde{x}_i, \tilde{x}_j) \leq (1 + \epsilon_d) \|\tilde{x}_i - \tilde{x}_j\|_2 \quad (92)$$

Let γ_{margin} define the minimum geometric separation threshold separating true relational neighbors from non-neighbors on the clean manifold. Let Δx represent the maximum bounded noise perturbation affecting the coordinates, such that $\|x_i - \tilde{x}_i\|_2 \leq \frac{1}{2} \|\Delta x\|_2$. For exact nearest-neighbor recovery, the maximum possible embedded distance of a true relational pair (undergoing noise contraction) must remain strictly lower than the minimum possible embedded distance of a non-relational pair. This separation requirement establishes the fundamental inequality:

$$(1 + \epsilon_d) (\gamma_{\text{margin}} - \|\Delta x\|_2) < (1 - \epsilon_d) (\gamma_{\text{margin}} + \|\Delta x\|_2) \quad (93)$$

We expand both sides of the inequality algebraically:

$$\gamma_{\text{margin}} - \|\Delta x\|_2 + \epsilon_d \gamma_{\text{margin}} - \epsilon_d \|\Delta x\|_2 < \gamma_{\text{margin}} + \|\Delta x\|_2 - \epsilon_d \gamma_{\text{margin}} - \epsilon_d \|\Delta x\|_2 \quad (94)$$

Subtracting γ_{margin} and adding $\epsilon_d \|\Delta x\|_2$ to both sides simplifies the expression to:

$$-\|\Delta x\|_2 + \epsilon_d \gamma_{\text{margin}} < \|\Delta x\|_2 - \epsilon_d \gamma_{\text{margin}} \quad (95)$$

Grouping the terms containing ϵ_d on the left-hand side and the noise terms on the right-hand side yields:

$$2\epsilon_d \gamma_{\text{margin}} < 2\|\Delta x\|_2 \implies \epsilon_d (\gamma_{\text{margin}} + \|\Delta x\|_2) \leq \gamma_{\text{margin}} - \|\Delta x\|_2 \quad (96)$$

Isolating the embedding distortion factor ϵ_d directly establishes the upper bound:

$$\epsilon_d \leq \frac{\gamma_{\text{margin}} - \|\Delta x\|_2}{\gamma_{\text{margin}} + \|\Delta x\|_2} \quad (97)$$

Enforcing this condition prevents any false-positive swaps in the nearest-neighbor topological map, ensuring exact geometric recovery of all omitted relational edges. $\square$

4.5 Theorem 4.5: Internal Verification Completeness

Theorem 4.5 (Internal Verification Completeness). *All symbolic validation checks are internally resolvable within the shadow-worker architecture without requiring external environmental reward feedback loops, achieving self-contained mathematical closure.*

Proof. Let $\mathcal{Y}$ define the symbolic codebook domain and $\mathcal{X}$ denote the continuous representation manifold. We formalize the projection pipeline via two mapping operators: the embedding operator $\text{Embed} : \mathcal{Y} \rightarrow \mathcal{X}$, and the active dictionary reconstruction operator $D_t : \mathcal{X} \rightarrow \mathcal{Y}$. By invoking the Dictionary Completeness property proven in Theorem 4.2, any recurrent discrete pattern maps completely into the active codebook domain, which establishes the strict image containment condition:

$$\text{Image}(\text{Embed}) \subseteq \text{Domain}(D_t) \quad (98)$$

The shadow-worker framework executes internal verification by evaluating the algebraic self-consistency of the composite loop mapping over internal coordinates. For an arbitrary symbolic input state $y \in \mathcal{Y}$, the reconstruction error norm is formulated as:

$$\mathcal{E}_{\text{internal}} = \|\text{Embed}(D_t(\text{Embed}(y))) - \text{Embed}(y)\|_2 \quad (99)$$

Because $\text{Image}(\text{Embed})$ is a subset of $\text{Domain}(D_t)$, the composite operator $\mathcal{T}_{\text{comp}} = \text{Embed} \circ D_t \circ \text{Embed}$ is well-defined and maps $\mathcal{Y} \rightarrow \mathcal{X}$. The evaluation of the norm $\mathcal{E}_{\text{internal}}$ requires access only to the current internal dictionary states D_t and the internal weights of the embedding layer. We check for algebraic consistency by verifying that $\mathcal{E}_{\text{internal}} \leq \epsilon_{\text{arith}}$, where ϵ_{arith} is the bounded machine precision parameter. Because this inequality depends entirely on internal parameters and coordinates, it can be computed without any external environmental reward function $R(x, a)$. This confirms that the validation routine achieves complete mathematical closure inside the shadow-worker framework. $\square$

4.6 Theorem 4.6: Explicit Hierarchical Error Bound

Theorem 4.6 (Explicit Hierarchical Error Bound). *The cascading failure probability P_{top} of a multi-level architectural stack satisfies the linear multiplicative upper bound:*

$$P_{\text{top}} \leq \sum_{\ell=1}^L p_{\ell} + p_{\text{src}} + \epsilon_{\text{arith}} \quad (100)$$

where p_{ℓ} is the failure probability at layer ℓ , p_{src} is the source mutation probability, and ϵ_{arith} is the internal validation error.

Proof. Let E_{top} represent the event that a top-level cascading failure occurs across an L -layer processing hierarchy. A top-level failure is triggered if any individual layer $\ell \in \{1, \dots, L\}$ fails (event E_{ℓ}), or if the underlying source data mutates destructively (event E_{src}), or if an internal arithmetic verification failure occurs (event E_{arith}). Expressing the top-level failure as a union of constituent failure events:

$$E_{\text{top}} = \left(\bigcup_{\ell=1}^L E_{\ell} \right) \cup E_{\text{src}} \cup E_{\text{arith}} \quad (101)$$

We evaluate the probability of system success, which corresponds to the joint intersection of the complements of all constituent failure events:

$$\mathbb{P}(E_{\text{top}}^c) = \mathbb{P} \left(\bigcap_{\ell=1}^L E_{\ell}^c \cap E_{\text{src}}^c \cap E_{\text{arith}}^c \right) \quad (102)$$

Assuming conditional independence or lower bounds on successive operational transitions, we expand the joint success probability as a product of conditional success probabilities:

$$\mathbb{P}(E_{\text{top}}^c) \geq \left(\prod_{\ell=1}^L (1 - p_{\ell}) \right) (1 - p_{\text{src}})(1 - \epsilon_{\text{arith}}) \quad (103)$$

where $p_\ell = \mathbb{P}(E_\ell)$, $p_{\text{src}} = \mathbb{P}(E_{\text{src}})$, and $\epsilon_{\text{arith}} = \mathbb{P}(E_{\text{arith}})$. We now apply the Weierstrass product inequality, which states that for any finite set of real numbers $a_i \in [0, 1]$:

$$\prod_i (1 - a_i) \geq 1 - \sum_i a_i \quad (104)$$

Applying this inequality to the bounded product of success probabilities yields:

$$\mathbb{P}(E_{\text{top}}^c) \geq 1 - \left(\sum_{\ell=1}^L p_\ell + p_{\text{src}} + \epsilon_{\text{arith}} \right) \quad (105)$$

Because the total probability space is normalized, the cascading failure probability is the complement of the success probability: $P_{\text{top}} = \mathbb{P}(E_{\text{top}}) = 1 - \mathbb{P}(E_{\text{top}}^c)$. Substituting this identity gives:

$$P_{\text{top}} \leq 1 - \left[1 - \left(\sum_{\ell=1}^L p_\ell + p_{\text{src}} + \epsilon_{\text{arith}} \right) \right] = \sum_{\ell=1}^L p_\ell + p_{\text{src}} + \epsilon_{\text{arith}} \quad (106)$$

This completes the formal derivation of the explicit linear upper bound. $\square$

4.7 Theorem 4.7: Hardware-Agnostic Realizability

Theorem 4.7 (Hardware-Agnostic Realizability). *A physical compute architecture guarantees stable real-time memory consolidation transitions without latency collapse if the parallel matrix-vector throughput R_{compute} and the memory bus bandwidth $B_{\text{bandwidth}}$ satisfy:*

$$R_{\text{compute}} \geq \frac{k \cdot d}{\tau_{S_{\text{max}}}}, \quad \text{and} \quad B_{\text{bandwidth}} \geq \frac{D_{\text{proto}}}{\tau_{L_{\text{max}}}} \quad (107)$$

Proof. We analyze the real-time execution constraints by decoupling the compute operations from the memory transfer tasks. **1. Computational Throughput Bound:** Each hierarchical spectral projection step requires projecting a d -dimensional input vector onto a k -dimensional sub-space spanned by the columns of the semi-orthogonal matrix $W \in \mathbb{R}^{d \times k}$. Mathematically, this operation is evaluated as the matrix-vector product $y = W^\top x$. Computing this product requires exactly $k \cdot d$ multiply-accumulate (MAC) operations. Let $\Delta\tau_{\text{compute}}$ denote the physical time required by the hardware processor to complete these operations. Given a hardware computational throughput of R_{compute} MAC operations per second, the processing latency evaluates to:

$$\Delta\tau_{\text{compute}} = \frac{k \cdot d}{R_{\text{compute}}} \quad (108)$$

To prevent real-time latency collapse, this processing time must not exceed the strict short-term operational latency budget $\tau_{S_{\text{max}}}$ allocated by the system clock:

$$\frac{k \cdot d}{R_{\text{compute}}} \leq \tau_{S_{\text{max}}} \implies R_{\text{compute}} \geq \frac{k \cdot d}{\tau_{S_{\text{max}}}} \quad (109)$$

2. Memory Bandwidth Bound: During the long-term consolidation phase, a consolidated prototype memory block of size D_{proto} bytes must be transferred across the physical memory bus. Let $\Delta\tau_{\text{bus}}$ denote the data transmission latency. Given a hardware bus bandwidth of $B_{\text{bandwidth}}$ bytes per second, the transmission time is:

$$\Delta\tau_{\text{bus}} = \frac{D_{\text{proto}}}{B_{\text{bandwidth}}} \quad (110)$$

To maintain structural tracking stability, this transmission must conclude within the long-term consolidation latency budget $\tau_{L_{\max}}$:

$$\frac{D_{\text{proto}}}{B_{\text{bandwidth}}} \leq \tau_{L_{\max}} \implies B_{\text{bandwidth}} \geq \frac{D_{\text{proto}}}{\tau_{L_{\max}}} \quad (111)$$

Simultaneous satisfaction of both inequalities ensures that the hardware can execute all operations within their respective timing budgets, precluding real-time latency collapse. $\square$

4.8 Lemma 4.8: Entropy Monotonicity

Lemma 4.8 (Entropy Monotonicity). *Under stable, contractive memory consolidation updates, the spectral von Neumann entropy $S(\rho)$ of the system density operator is strictly monotonic and non-increasing over time: $S(\rho_{t+1}) \leq S(\rho_t)$.*

Proof. Let ρ_t represent the symmetric, positive semi-definite system density operator at time step t , normalized such that $\text{Tr}(\rho_t) = 1$. The von Neumann entropy is defined as:

$$S(\rho_t) = -\text{Tr}(\rho_t \log \rho_t) = -\sum_{i=1}^d \lambda_i(t) \log \lambda_i(t) \quad (112)$$

where $\lambda(t) = [\lambda_1(t), \lambda_2(t), \dots, \lambda_d(t)]^\top$ is the vector of sorted eigenvalues of ρ_t . Under stable contractive consolidation, the transformation of the density operator can be modeled as a trace-preserving completely positive map. This restricts the evolution of the clean eigenvalue spectrum to a random walk governed by a doubly stochastic mixing matrix $M \in \mathbb{R}^{d \times d}$ (where $\sum_i M_{ij} = 1$ and $\sum_j M_{ij} = 1$):

$$\lambda(t+1) = M\lambda(t) \quad (113)$$

By the fundamental properties of majorization, because M is a doubly stochastic matrix, the eigenvalue vector at time step t majorizes the updated eigenvalue vector at step $t+1$, which is written notationally as:

$$\lambda(t) \prec \lambda(t+1) \quad (114)$$

The continuous function $H(x) = -\sum x_i \log x_i$ is strictly Schur-concave over the unit simplex. By the defining criterion of Schur-concavity, if a vector x is majorized by a vector y ($x \prec y$), the functional value satisfies $H(x) \geq H(y)$. Mapping this property directly to our majorized eigenvalue spectra:

$$\lambda(t) \prec \lambda(t+1) \implies H(\lambda(t+1)) \leq H(\lambda(t)) \quad (115)$$

Substituting the definition of the von Neumann entropy $S(\rho_t) = H(\lambda(t))$ into the inequality yields:

$$S(\rho_{t+1}) \leq S(\rho_t) \quad (116)$$

This mathematically demonstrates that the spectral entropy is strictly monotonic and non-increasing under contractive consolidation updates. $\square$

4.9 Lemma 4.9: GIM Mandatory Necessity

Lemma 4.9 (GIM Mandatory Necessity). *Operating across non-isomorphic state space manifolds ($\mathcal{M}_1 \not\cong \mathcal{M}_2 \not\cong \mathcal{M}_3$) requires a decoupled global integrity monitor to prevent boundary tracking drift and system-wide trajectory collapse.*

Proof. Let $\mathcal{M}_1, \mathcal{M}_2$, and $\mathcal{M}_3$ represent distinct topological manifolds that host different state components of the model architecture. Because these spaces are non-isomorphic ($\mathcal{M}_i \not\cong \mathcal{M}_j$ for $i \neq j$), there exists no global coordinate chart, bijective homeomorphism, or continuous isometric mapping $\phi : \mathcal{M}_i \rightarrow \mathcal{M}_j$ capable of matching their metric tensors. Let $\nabla_{\mathcal{M}_i} E_i$ denote a localized stabilizer operating solely within the tangent space $T_x \mathcal{M}_i$ of manifold $\mathcal{M}_i$. Because the local stabilizer is restricted to $T_x \mathcal{M}_i$, its orthogonal projection operator satisfies:

$$\langle \nabla_{\mathcal{M}_i} E_i, v \rangle = 0, \quad \forall v \in T_x \mathcal{M}_j \quad (j \neq i) \quad (117)$$

Consequently, a purely local tracking system cannot sense or counteract cross-manifold boundary drift. Unmonitored cross-coupling terms can introduce energy variations that escape the local projections, causing tracking errors to accumulate over time. To guarantee non-divergence, we define a decoupled global monitor that evaluates a scalar Lyapunov functional V_{GIM} over the joint product manifold $\mathcal{M}_{\text{joint}} = \mathcal{M}_1 \times \mathcal{M}_2 \times \mathcal{M}_3$:

$$V_{\text{GIM}}(x) = \sum_{i=1}^3 V_i(x^{(i)}) + \sum_{i \neq j} \Psi_{ij}(x^{(i)}, x^{(j)}) \quad (118)$$

where Ψ_{ij} represents the interaction cross-terms. Differentiating V_{GIM} along the joint trajectory balances the cross-manifold variations explicitly. Since local stabilizers cannot evaluate the interaction terms Ψ_{ij} due to the lack of an isomorphic mapping, a decoupled global monitor is topologically mandatory to prevent tracking collapse. $\square$

4.10 Lemma 4.10: Spectral Drift Suppression Bound

Lemma 4.10 (Spectral Drift Suppression Bound). *The stochastic approximation tracking updates executed on the Stiefel manifold are strictly bounded in Frobenius norm by:*

$$\left\| \hat{W}_{t+1} - \hat{W}_t \right\|_F \leq \eta_t B^2 \quad (119)$$

where η_t is the step-size and B is the sub-Gaussian activation bound.

Proof. The discrete-time tracking update for the semi-orthogonal projection matrix $\hat{W}_t \in \mathcal{V}_k(\mathbb{R}^d)$ on the Stiefel manifold is governed by the Oja update equation:

$$\hat{W}_{t+1} - \hat{W}_t = \eta_t \left(I_d - \hat{W}_t \hat{W}_t^\top \right) x_t x_t^\top \hat{W}_t \quad (120)$$

We define the step increment matrix as $\Delta \hat{W}_t = \hat{W}_{t+1} - \hat{W}_t$. Taking the Frobenius norm of both sides yields:

$$\left\| \Delta \hat{W}_t \right\|_F = \eta_t \left\| \left(I_d - \hat{W}_t \hat{W}_t^\top \right) x_t x_t^\top \hat{W}_t \right\|_F \quad (121)$$

We rewrite the matrix product inside the norm by grouping terms into a column vector $u \in \mathbb{R}^d$ and a row vector $v^\top \in \mathbb{R}^{1 \times k}$:

$$u = \left(I_d - \hat{W}_t \hat{W}_t^\top \right) x_t, \quad \text{and} \quad v^\top = x_t^\top \hat{W}_t \quad (122)$$

This expresses the product as a rank-one matrix: $\left(I_d - \hat{W}_t \hat{W}_t^\top \right) x_t x_t^\top \hat{W}_t = uv^\top$. The Frobenius norm of any rank-one outer product matrix simplifies exactly to the product of the Euclidean norms of its constituent vectors:

$$\left\| uv^\top \right\|_F = \|u\|_2 \|v\|_2 \quad (123)$$

We bound the Euclidean norm of vector u using the properties of orthogonal projectors. Because $\hat{W}_t$ resides on the Stiefel manifold, it satisfies $\hat{W}_t^\top \hat{W}_t = I_k$. This implies that $P = \hat{W}_t \hat{W}_t^\top$ is an orthogonal projection matrix, meaning its orthogonal complement $P_\perp = I_d - \hat{W}_t \hat{W}_t^\top$ is also an orthogonal projection matrix with a spectral norm of exactly $\|I_d - \hat{W}_t \hat{W}_t^\top\|_2 = 1$. Consequently:

$$\|u\|_2 = \left\| \left(I_d - \hat{W}_t \hat{W}_t^\top \right) x_t \right\|_2 \leq \left\| I_d - \hat{W}_t \hat{W}_t^\top \right\|_2 \|x_t\|_2 \leq 1 \cdot B = B \quad (124)$$

where $\|x_t\|_2 \leq B$ represents the uniform sub-Gaussian bound on the activation vectors. Next, we bound the Euclidean norm of the row vector $v^\top$. Using the sub-multiplicative property of the spectral norm:

$$\|v\|_2 = \left\| \hat{W}_t^\top x_t \right\|_2 \leq \left\| \hat{W}_t^\top \right\|_2 \|x_t\|_2 \leq 1 \cdot B = B \quad (125)$$

since the spectral norm of any semi-orthogonal matrix is exactly $\|\hat{W}_t\|_2 = 1$. Substituting the vector norm bounds from (124) and (125) back into the rank-one Frobenius identity in (123) yields:

$$\left\| \Delta \hat{W}_t \right\|_F = \eta_t \|u\|_2 \|v\|_2 \leq \eta_t \cdot B \cdot B = \eta_t B^2 \quad (126)$$

This completes the rigorous proof of the spectral drift suppression bound. $\square$

4.11 Lemma 4.11: Frequency-Type Partitioning Rule

Lemma 4.11 (Frequency-Type Partitioning Rule). *The Type-Conditioned Spectral Sorter maps activation vectors into orthogonal complements, ensuring that $\langle x^{(i)}, x^{(j)} \rangle = 0$ for all distinct type indices $i \neq j$.*

Proof. Let $x \in \mathbb{R}^d$ denote the global activation vector. The type-conditioned activation components $x^{(i)}$ and $x^{(j)}$ are generated by projecting x onto distinct subspaces using orthogonal projection operators:

$$x^{(i)} = \Pi_{V_i} x = V_i V_i^\top x, \quad \text{and} \quad x^{(j)} = \Pi_{V_j} x = V_j V_j^\top x \quad (127)$$

where $V_i \in \mathbb{R}^{d \times k_i}$ and $V_j \in \mathbb{R}^{d \times k_j}$ represent the orthonormal basis matrices for the i -th and j -th type sub-manifolds, respectively. We evaluate the inner product between these two type-conditioned activations:

$$\langle x^{(i)}, x^{(j)} \rangle = \left(x^{(i)} \right)^\top x^{(j)} = \left(V_i V_i^\top x \right)^\top \left(V_j V_j^\top x \right) \quad (128)$$

Applying the matrix transposition rule $(AB)^\top = B^\top A^\top$ and expanding the product yields:

$$\langle x^{(i)}, x^{(j)} \rangle = x^\top V_i \left(V_i^\top V_j \right) V_j^\top x \quad (129)$$

By architectural design, the type-conditioned spectral sorter enforces mutual orthogonality between different type bases in the spectral domain. This requires the cross-product of the basis matrices to evaluate to the zero matrix for any $i \neq j$:

$$V_i^\top V_j = \mathbf{0}_{k_i \times k_j} \quad (130)$$

Substituting this null matrix back into the inner product expression yields:

$$\langle x^{(i)}, x^{(j)} \rangle = x^\top V_i \left(\mathbf{0}_{k_i \times k_j} \right) V_j^\top x = 0 \quad (131)$$

. This confirms that the type-conditioned activation components reside in orthogonal complements. $\square$

4.12 Lemma 4.12: Intrinsic Spectral Drift Detector Stability

Lemma 4.12 (Intrinsic Spectral Drift Detector Stability). *The empirical drift statistic evaluated using principal subspace angles bounds the tracking variance against population parameters at an asymptotic decay rate of $\mathcal{O}\left(\frac{1}{t\delta_k^2}\right)$.*

Proof. Let $V \in \mathcal{V}_k(\mathbb{R}^d)$ define the true population principal subspace matrix derived from the clean covariance matrix C . Let $\hat{V}_t \in \mathcal{V}_k(\mathbb{R}^d)$ denote the empirical principal subspace matrix computed from the sample covariance matrix $\hat{C}_t$ at step t . We quantify the geometric rotation between the empirical and population subspaces using the Frobenius norm of the sine of their principal angles, $\left\|\sin \Theta(\hat{V}_t, V)\right\|_F$. By the Davis-Kahan sin Θ Theorem, this angular deviation is upper-bounded by the Frobenius norm of the covariance perturbation divided by the population spectral eigengap $\delta_k = \lambda_k - \lambda_{k+1} > 0$:

$$\left\|\sin \Theta(\hat{V}_t, V)\right\|_F \leq \frac{\left\|\hat{C}_t - C\right\|_F}{\delta_k} \quad (132)$$

Squaring both sides and taking the total expectation yields:

$$\mathbb{E} \left[\left\|\sin \Theta(\hat{V}_t, V)\right\|_F^2 \right] \leq \frac{\mathbb{E} \left[\left\|\hat{C}_t - C\right\|_F^2 \right]}{\delta_k^2} \quad (133)$$

We analyze the numerator by modeling the empirical covariance matrix as a sum of independent, rank-one sample updates: $\hat{C}_t = \frac{1}{t} \sum_{\tau=1}^t x_\tau x_\tau^\top$. Applying the Matrix Bernstein inequality to this sum of independent sub-Gaussian random matrices bounds the mean squared error of the empirical covariance estimation:

$$\mathbb{E} \left[\left\|\hat{C}_t - C\right\|_F^2 \right] \leq \frac{c_0}{t} \quad (134)$$

where $c_0 > 0$ is a constant scaling factor determined by the sub-Gaussian norm of the activation distribution. Substituting the estimation bound from (134) directly into the Davis-Kahan expectation inequality in (133) yields:

$$\mathbb{E} \left[\left\|\sin \Theta(\hat{V}_t, V)\right\|_F^2 \right] \leq \frac{c_0}{t\delta_k^2} = \mathcal{O} \left(\frac{1}{t\delta_k^2} \right) \quad (135)$$

This confirms that the variance of the empirical drift statistic decays to zero at an asymptotic rate of $\mathcal{O}(1/t)$, isolating true structural drift from isotropic statistical noise. $\square$

4.13 Theorem 4.13: Analytic Boundary Patch

Theorem 4.13 (Analytic Boundary Patch). *Applying Topological Ridge Regularization to a density operator shifts its spectrum away from boundary branch points, restoring an exponential Chebyshev approximation convergence rate of $\mathcal{O}(e^{-M\chi})$ for expansion filters over non-analytic spectral singularities.*

Proof. The unregularized spectral log-volume and von Neumann entropy functionals require evaluating terms of the form $f(x) = -x \ln x$ or $f(x) = \log x$ on the eigenvalues of the system density operator $\hat{\rho}$. The derivative of this functional, $f'(x) = -1 - \ln x$, diverges to infinity as $x \rightarrow 0^+$. This indicates the presence of a branch point singularity at the origin of the complex plane, which lies on the boundary of the active operational spectrum $[0, 1]$. When expanding $f(x)$ using a polynomial

basis, this boundary singularity slows the approximation truncation error convergence from an exponential rate to an algebraic rate of $\mathcal{O}(1/M)$. Topological Ridge Regularization resolves this by adding an explicit algorithmic precision floor $\delta > 0$ to the density operator, defining the patched operator $\tilde{\rho}$:

$$\tilde{\rho} = \hat{\rho} + \delta I_d \quad (136)$$

This transformation shifts every eigenvalue λ_i of the density operator to $\tilde{\lambda}_i = \lambda_i + \delta$. Because the original eigenvalues are non-negative ($\lambda_i \geq 0$), the regularized spectrum is strictly restricted to the shifted interval:

$$\tilde{\lambda}_i \in [\delta, 1 + \delta] \quad (137)$$

This shift moves the operating domain away from the branch point singularity at $x = 0$. We can now construct a Bernstein ellipse E_ρ in the complex plane, centered around the regularized support interval $[\delta, 1 + \delta]$. The semi-minor axis length of this ellipse is chosen to be small enough so that it does not intersect the singularity at the origin:

$$\text{dist}(0, E_\rho) > 0 \quad (138)$$

Because the shifted function $f(z)$ is strictly holomorphic (analytic) within the interior of this Bernstein ellipse, Chebyshev expansion theory guarantees that the truncation error of an M -th order polynomial expansion filter decays exponentially:

$$\max_{z \in [\delta, 1 + \delta]} |f(z) - p_M(z)| \leq \frac{2M_f}{\rho^M - 1} = \mathcal{O}(e^{-M\chi}) \quad (139)$$

where $\chi = \ln \rho > 0$ represents the ellipse parameter and M_f bounds the functional text inside the ellipse. This confirms that the regularization successfully restores exponential convergence. $\square$

4.14 Theorem 4.14: Global iISS Loop Stability

Theorem 4.14 (Global iISS Loop Stability). *The interconnected multi-variable feedback loop combining neural learning steps with hierarchical projection steps is globally Integral Input-to-State Stable (iISS) under sub-Gaussian activation bounds.*

Proof. We prove stability by constructing a composite global Lyapunov function using small-gain synthesis and analyzing its behavior under an exponential loss dissipation operator. Let $V_i(x_i)$ represent the local Input-to-State Stable (ISS) Lyapunov function for the i -th subsystem. We construct the composite global Lyapunov function $V(x)$ as a weighted sum of these local functions:

$$V(x) = \sum_{i=1}^m \sigma_i(V_i(x_i)) \quad (140)$$

where each σ_i is a smooth, strictly increasing scaling function. Differentiating $V(x)$ along the coupled system trajectories yields the following inequality:

$$\dot{V} \leq -\alpha_0(\|x\|_2) + \gamma_0(\|d_t\|_2) \quad (141)$$

where α_0 is a continuous positive definite function, γ_0 is a $\mathcal{K}_\infty$ gain function, and d_t represents the external tracking disturbance vector. To prevent transient or persistent adversarial inputs from driving the tracking bounds to infinity (the Supremum Trap), the architecture embeds an Exponential Loss Dissipation Operator by adding an auxiliary fading-memory variable W_t :

$$W_t = \int_0^t e^{-\lambda(t-\tau)} \gamma_0(\|d_\tau\|_2) d\tau \quad (142)$$

where $\lambda > 0$ represents the engineered dissipation rate parameter. Differentiating W_t with respect to time yields the linear evolution filter equation:

$$\dot{W}_t = -\lambda W_t + \gamma_0 (\|d_t\|_2) \quad (143)$$

We formulate the modified global integrity functional $V_{\text{GIM}}(x, W)$ as:

$$V_{\text{GIM}}(x, W) = V(x) + W_t \quad (144)$$

Evaluating the total time derivative of this modified global functional:

$$\begin{aligned} \dot{V}_{\text{GIM}} &= \dot{V} + \dot{W}_t \\ &\leq [-\alpha_0 (\|x\|_2) + \gamma_0 (\|d_t\|_2)] + [-\lambda W_t + \gamma_0 (\|d_t\|_2)] \\ &= -\alpha_0 (\|x\|_2) - \lambda W_t + 2\gamma_0 (\|d_t\|_2) \end{aligned} \quad (145)$$

Integrating this derivative equation from 0 to t establishes the explicit trajectory tracking bound:

$$\alpha (\|x_t\|_2) \leq \beta (\|x_0\|_2, t) + \int_0^t e^{-\lambda(t-\tau)} \gamma_{\text{new}} (\|d_\tau\|_2) d\tau \quad (146)$$

Because the composite small-gain contraction condition $\Gamma_{\text{loop}}(s) < s$ is enforced via algorithmic step-size normalization, tracking perturbations are proven to decay asymptotically to zero, completing the proof. $\square$